Einstein's equation, prior to my trip down Interstate 5 in 1985, was suitable for Carl Sagan's novel, because none of them could be traversed safely. Each and everyone of them was predicted to evolve with time in a very peculiar way: The wormhole is created at some moment of time, opens up briefly, and then pinches off and disappears—and its total life span from creation to pinch-off is so short that nothing whatsoever (no person, no radiation, no signal of any sort) can travel through it, from one mouth to the other. Anything that tries will get caught and destroyed in the pinch-off. Figure 14.2 shows a simple example.

Like most of my physicist colleagues, I have been skeptical of wormholes for decades. Not only does Einstein's field equation predict that wormholes live short lives if left to their own devices; their lives are made even shorter by random infalling bits of radiation: The radiation (according to calculations by Doug Eardley and Ian Redmount) gets accelerated to ultra-high energy by the wormhole's gravity, and as the energized radiation bombards the wormhole's throat, it triggers the throat to recontract and pinch off far faster than it would otherwise—so fast, in fact, that the wormhole has hardly any life at all.

There is another reason for skepticism. Whereas *black holes* are an inevitable consequence of stellar evolution (massive, slowly spinning stars, of just the sort that astronomers see in profusion in our galaxy, will implode to form black holes when they die), there is no analogous, natural way for a *wormhole* to be created. In fact, there is no reason at all to think that our Universe contains today *any* singularities of the sort that give birth to wormholes (Figure 14.2); and even if such singularities did exist, it is hard to understand how two of them could find each other in the vast reaches of hyperspace, so as to create a wormhole in the manner of Figure 14.2.

When one's friend needs help, one is willing to turn most anywhere that help might be found. Wormholes—despite my skepticism about them—seemed to be the only help in sight. Perhaps, it occurred to me on Interstate 5 somewhere west of Fresno, there is some way that an infinitely advanced civilization could hold a wormhole open, that is, prevent it from pinching off, so that Eleanor Arroway could travel through it from Earth to Vega and back. I pulled out pen and paper and began to calculate. (Fortunately, Interstate 5 is very straight; I could calculate without getting carsick.)